SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Department of Artificial Intelligence and Machine Learning

DATA VISUALIZATION

Integrating R with Power BI for Advanced Analytics

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1. Introduction

Business intelligence becomes more powerful when interactive dashboards are combined with statistical analysis. Power BI provides KPI cards, slicers, drill-down and native charts, while R adds flexible statistical analysis and visualizations. In this practical, the existing Adventure Works sales model is extended with R visuals to explore historical sales, seasonality and forecasting.

2. Aim and Objectives

Aim: To integrate R programming with Power BI and use R-based visualizations for advanced analysis of the Adventure Works dataset.

- Configure R scripting in Power BI Desktop.
- Reuse the existing data model and existing DAX measures without creating new measures for R visuals.
- Build R visuals for historical trends, forecasting, seasonality and forecast validation.
- Combine R graphs with native Power BI charts, KPI cards and slicers.
- Interpret results using actual dashboard output and acknowledge forecasting uncertainty.

3. Dataset and Tools

Dataset: Adventure Works, using the existing sales table and related lookup tables such as Product Lookup, Calendar Lookup and Territory/Region. Existing measures include Total Sales, Total Profit, Total Orders, Total Customers, Total Quantity, Average Sales, Average Order Value, Profit Margin %, Sales Previous Month, Sales Variance and Sales Growth %. Tools used: Power BI Desktop, R for Windows, Power Query Editor for prior data preparation, existing DAX measures and Power BI R visuals.

4. Dashboard and KPI Cards

The dashboard displays six KPI cards: Total Sales (24.91M), Total Profit (10.46M), Total Orders (25K), Total Customers (17K), Total Quantity (84K), and Profit Margin (0.42). If Profit Margin is a decimal ratio, 0.42 corresponds to approximately 42% when formatted as a percentage. The cards provide a compact overview of revenue, profitability, order volume, customer reach and quantity sold. They respond to the report's filter context.

5. Native Power BI Visuals and Filters

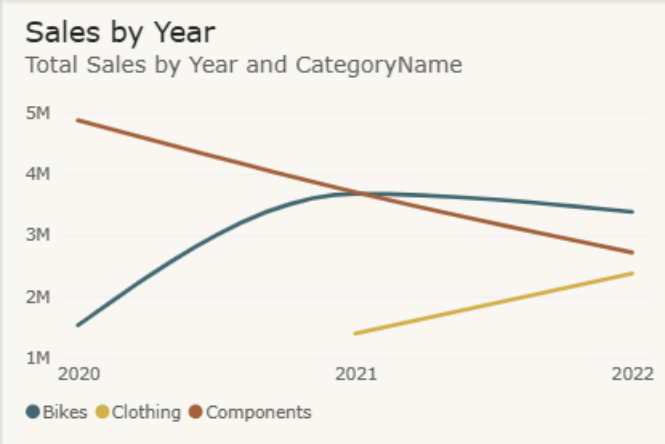
5.1 Sales by Category

The horizontal bar chart compares sales across product categories. In the supplied screenshot, Components has the highest displayed sales among the visible categories, followed by Bikes and Clothing. Horizontal bars make category comparisons easy to read.



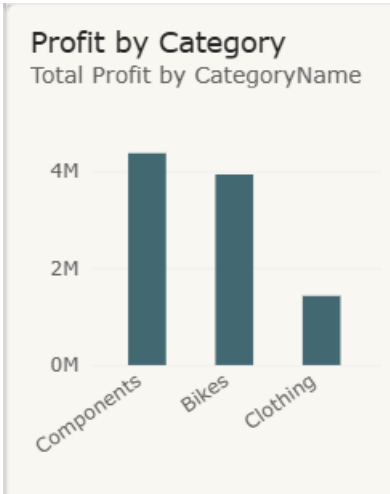
5.2 Sales by Year

The line chart shows yearly sales movement. The displayed series rises from 2020 to 2021 and then decreases slightly in 2022. Since this chart is yearly, it does not establish which individual month had the highest or lowest sales.



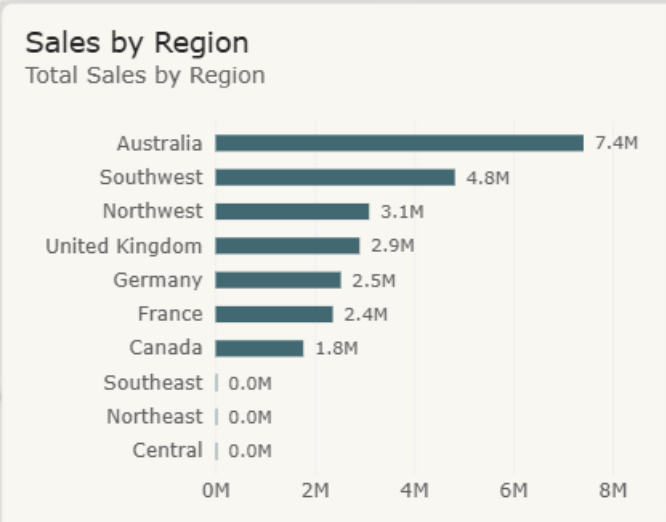
5.3 Profit by Category

The column chart compares profit across product categories. Components has the highest displayed profit among the categories shown, followed by Bikes and Clothing.



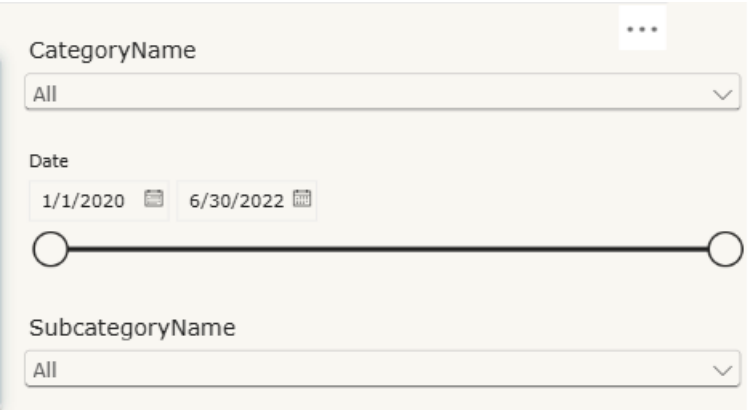
5.4 Sales by Region

The regional bar chart compares sales across territories. Australia has the longest displayed bar in the supplied screenshot, followed by Southwest and Northwest among the visible regions.



5.5 Slicers and Interactivity

The screenshot includes slicers for CategoryName, Date and SubcategoryName. They allow the report to be narrowed to selected categories, date ranges and subcategories. Slicers should be tested to confirm that the relevant native charts respond to selections.



6. R Integration Procedure

Step 1 — Install and configure R

R was installed on Windows and configured in Power BI Desktop under File → Options and settings → Options → R scripting.

Step 2 — Add an R visual

An R visual was added from the Visualizations pane. The required existing columns and measures were placed in the Values field well.

Step 3 — Pass fields to R

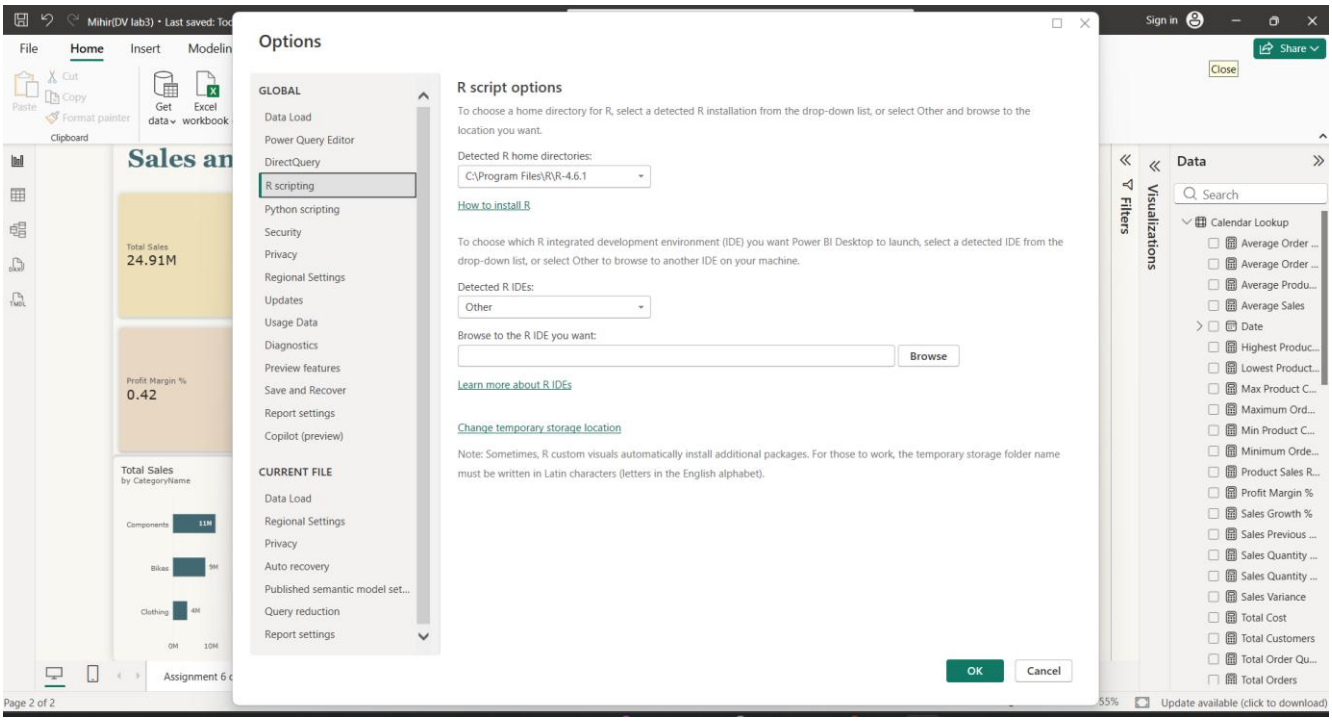
Power BI passes the selected fields to the script in a data frame named dataset. The scripts use the columns supplied to the visual; field order must match the script's assumptions.

Step 4 — Prepare data

The scripts remove missing dates/categories and numeric values, sort time-based data chronologically, and group supplied values for category or calendar-month comparisons where appropriate.

Step 5 — Run and verify

Each script is executed from the R visual editor. The chart is checked for readable axes, sensible values and compatibility with the existing Power BI visual results.



7. R Visualizations

7.1 Historical Sales Trend

This line graph plots an existing numeric sales measure against a date column. It makes increases, decreases and fluctuations easier to observe than a single KPI. No new DAX measure is created.

7.2 Sales Forecast

The forecasting visual uses Holt–Winters exponential smoothing from base R to estimate future values from the monthly series. With sufficient history, a seasonal model can be fitted. Forecast lines and prediction limits communicate the estimate and its uncertainty. Forecasts are not guaranteed outcomes.

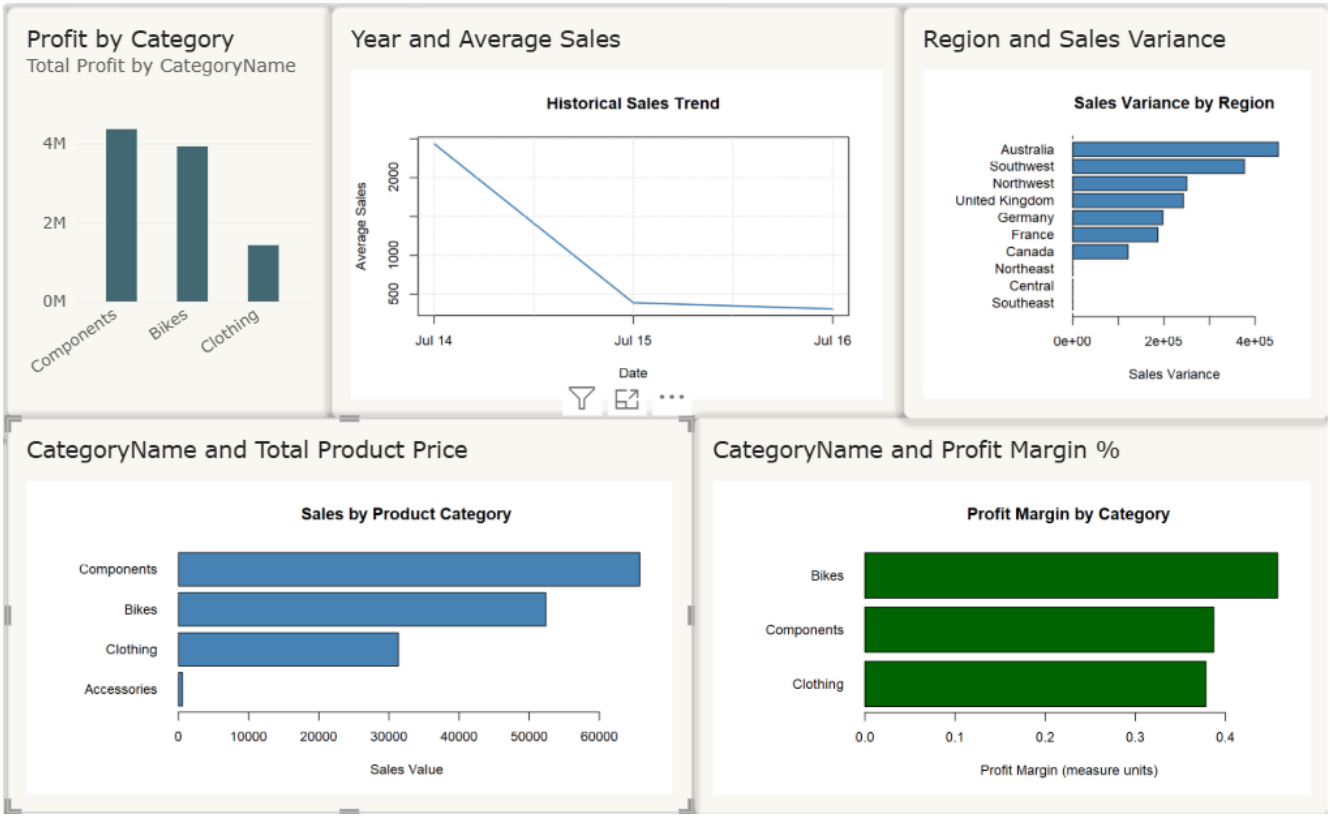
7.3 Average Sales by Calendar Month

The seasonality visual groups values by calendar month and plots their average. It helps explore whether some months tend to have higher or lower sales. The conclusion is more reliable when several complete years are available.

7.4 Forecast Validation: Actual vs Predicted

The validation visual reserves recent observations for testing, fits the model on earlier observations and compares predictions with actual values. Mean Absolute Error (MAE) is the average absolute prediction error. Root Mean Squared Error (RMSE) penalizes larger errors more heavily. Lower values are generally preferable when comparing models on the same data and test period.

R visuals render their plots as images. They may refresh when their input data or filter context changes, but the plotted marks do not provide the same cross-filtering interaction as native Power BI charts.



R script editor

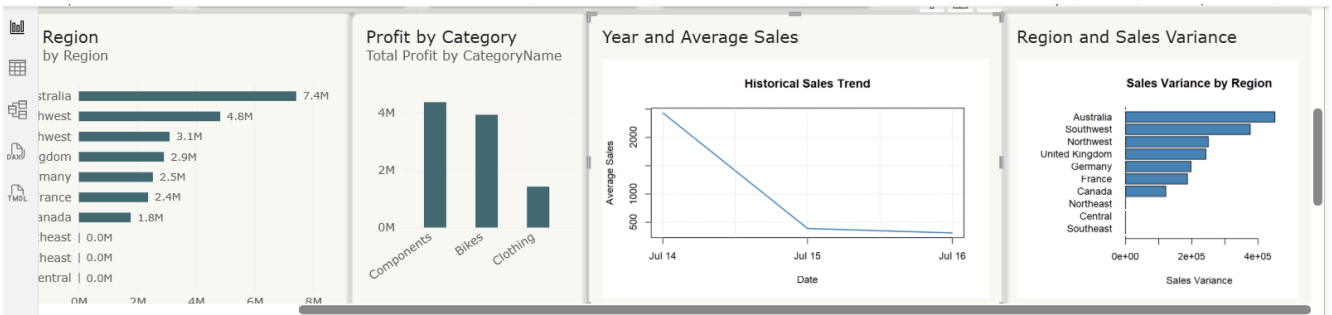
```r
df <- dataset
category <- as.character(df[[1]])
sales <- as.numeric(df[[2]])

ok <- !is.na(category) & !is.na(sales)
category <- category[ok]
sales <- sales[ok]

totals <- aggregate(sales, by = list(Category = category),
                    FUN = sum, na.rm = TRUE)
totals <- totals[order(totals$x), ]

par(mar = c(5, 9, 4, 2))
barplot(
```

Assignment 6 dashboard | Assignment 7 Dashboard +



R script editor

```r
# The following code to create a dataframe and remove duplicated rows is always executed and acts as a preamble for your script:

# dataset <- data.frame(Year, Quarter, Month, Day, Average Sales)
# dataset <- unique(dataset)

# Paste or type your script code here:
# R Visual 1: Historical Sales Trend

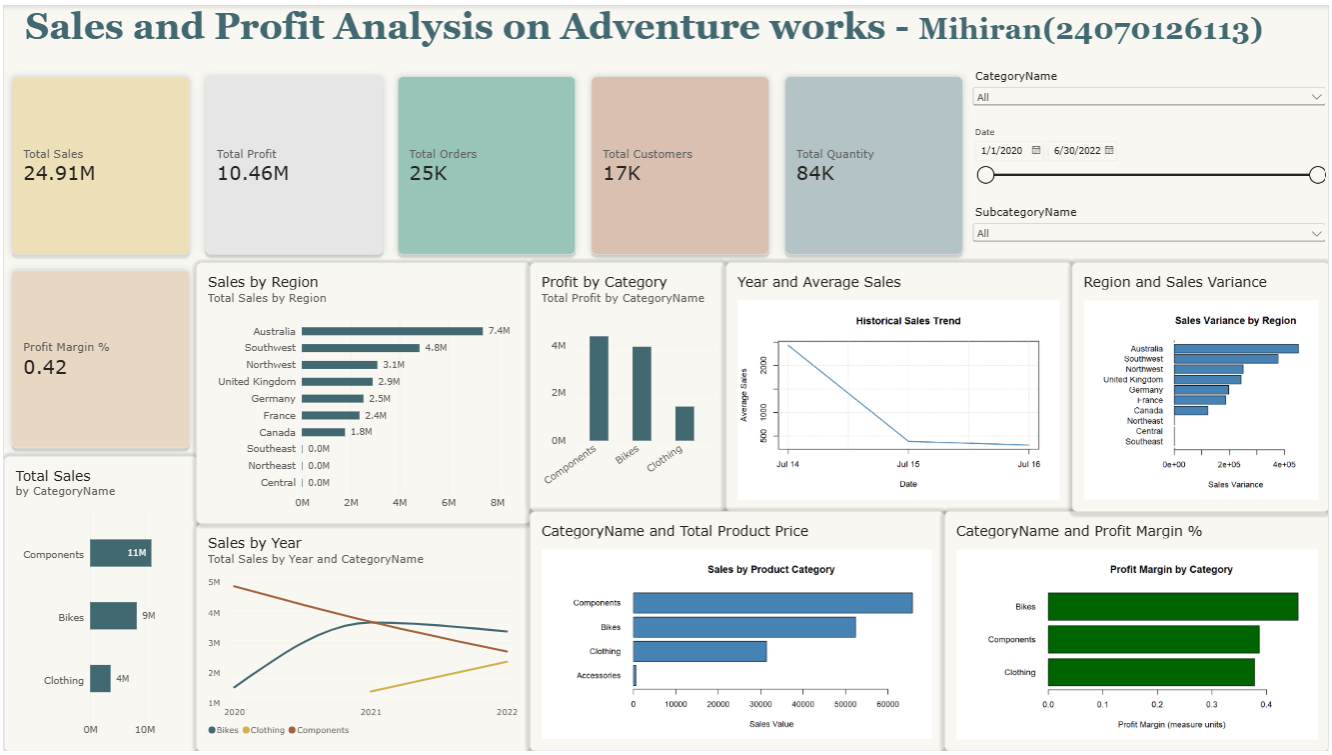
df <- dataset

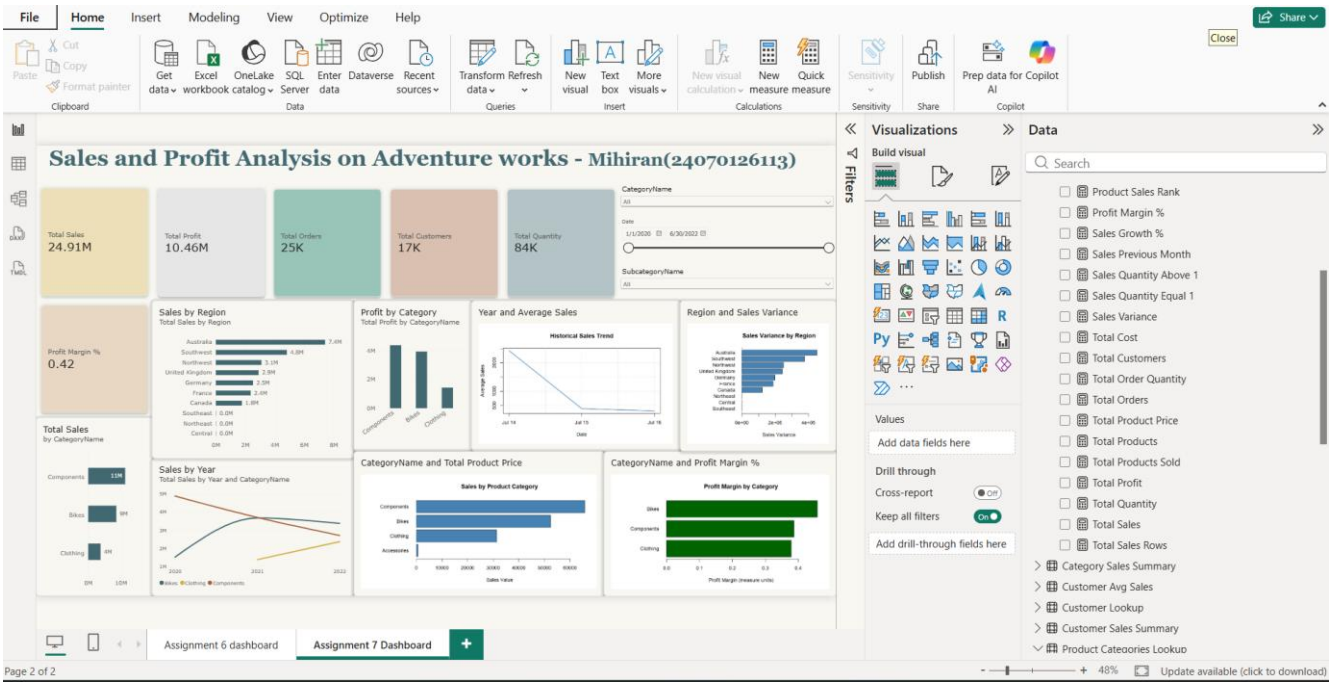
dates <- as.Date(df[[1]])
sales <- as.numeric(df[[2]])

ok <- !is.na(dates) & !is.na(sales)
dates <- dates[ok]
```

Assignment 6 dashboard | Assignment 7 Dashboard +

Final Dashboard Output





9. Results and Interpretation

From the supplied dashboard screenshot, Total Sales is 24.91M, Total Profit is 10.46M, Total Orders is 25K, Total Customers is 17K, Total Quantity is 84K and Profit Margin is 0.42. Components leads the displayed categories for sales and profit. Australia has the longest displayed regional sales bar. The yearly chart rises from 2020 to 2021 and then declines slightly in 2022.

Record the actual R forecast and validation results after running the scripts. Include the forecast horizon, visible pattern and MAE/RMSE values. Do not claim a specific best month or high forecast accuracy unless the generated output supports it.

10. Business Relevance

Category and regional charts help identify sales concentration. KPI cards summarise overall performance, while time-based analysis helps explore changes in demand. Forecasting may support inventory, staffing and promotion planning, but model error should be considered before using predictions for decisions.

11. Conclusion

This practical integrates R scripting with Power BI to extend the existing Adventure Works dashboard. Native Power BI visuals provide category, region and yearly comparisons, while R adds statistical plotting, forecasting, calendar-month analysis and holdout validation. Existing measures are reused rather than duplicated. Conclusions should be based on the actual results and should acknowledge forecast uncertainty and the interaction limitations of R visuals.